

Algorithmic Pseudocode Sample 18

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Inputs: s, i, d, a, r, e, v, h , daily vaccinations

Output: $\vec{\beta}_{uu}, \vec{\beta}_{vu}, \vec{\beta}_{vv}, \vec{\beta}_{uv}$

Initialization: $n = 7$ or $n = 14$

```
1: for each time step  $j$  do
2:   Select window  $z_j = \{j - n + 1, \dots, j\}$ 
3:   Initialize parameters  $\beta_{uu}, \beta_{vu}, \beta_{vv}, \beta_{uv}$ 
4:   Calculate the initial cost  $C_j(n, i, d, \hat{i}_j, \hat{d}_j)$ 
5:    $flag = 0$ 
6:   while  $flag = 0$  do
7:     Create the trial parameters set  $P_k$ 
8:     Calculate a cost using every parameter from  $P_k$  set
9:     Find the minimum of all costs  $C'_j(n, i, d, \hat{i}_j, \hat{d}_j)$ 
10:    if  $C'_j(n, i, d, \hat{i}_j, \hat{d}_j) < C_j(n, i, d, \hat{i}_j, \hat{d}_j)$  then
11:       $C_j(n, i, d, \hat{i}_j, \hat{d}_j) = C'_j(n, i, d, \hat{i}_j, \hat{d}_j)$ 
12:      Keep the modified infection rate and create new set of trial parameters  $P'_k$ 
13:    else
14:       $flag = 1$ 
15:       $\vec{\beta}_{uu}(j) = \beta_{uu}$ 
16:       $\vec{\beta}_{vu}(j) = \beta_{vu}$ 
17:       $\vec{\beta}_{vv}(j) = \beta_{vv}$ 
18:       $\vec{\beta}_{uv}(j) = \beta_{uv}$ 
19:    end if
20:  end while
21: end for
```
